

# THE CLOCKWISE PRIOR: SILENT, DIRECTED CONVENTION REVERSION IN LANGUAGE MODELS

Anonymous authors

Paper under double-blind review

## ABSTRACT

Many quantities in science and engineering carry a sign that is fixed only by convention: a mesh current is positive if it circulates clockwise, a line integral is positive if the curve is traversed counter-clockwise, a cross product follows the right hand. Both choices are always physically valid, so an instruction to use the non-standard choice is legitimate, unambiguous, and easy to state. We study what language models do with such instructions. Across three physics and mathematics domains whose answers are verified to the sign by four independent solvers, eight instructable conventions, and fifteen model configurations, we find that when models get these signs wrong, they do not fail randomly. Among 58,244 graded variables, 99.87% of all sign errors reproduce, exactly, the value that the *trained* convention would give; on the seven conventions whose transform is a pure sign flip, the figure is 100%. We call this failure mode *reversion*, and we map what governs it. First, reversion tracks how the convention’s reversal is taught: the one convention whose reversal is absent from textbooks (counter-clockwise mesh currents) defeats every frontier model we test (76.0% reversion), while conventions whose reversal is itself part of the curriculum are obeyed ( $\leq 1.4\%$ ). Second, task difficulty reactivates the prior: within a domain that models obey when problems are easy, hardening the problems raises reversion from 1.0% to 35.8%, while a control instruction demanding the identical output change stays below 1%. Third, compliance is capability-gated: mid-scale models revert on conventions frontier models obey. Mitigations help less than one would hope. Enabling a reasoning mode raises accuracy dramatically while leaving reversion nearly unchanged on most conventions, restoring compliance only where the reversal is both taught and procedurally shallow; and placing fully worked examples of the required convention in the prompt leaves reversion essentially unchanged in two of three frontier models. Two predictions we pre-registered were refuted by the data, and we report both. The benchmark regenerates deterministically from seeds and uses no language-model judges anywhere.

## 1 INTRODUCTION

Ask a language model to analyze a simple circuit, but add one sentence: “mesh currents are defined counter-clockwise.” Circuit textbooks almost universally draw mesh currents clockwise, but nothing in physics prefers one direction; the choice only fixes which currents are reported as positive. A model that follows the instruction produces the same magnitudes as usual with certain signs flipped. What we observe, across every capable model we test, is different: the model sets up correct equations, solves them correctly, recovers every magnitude, and then reports the signs that a *clockwise* analysis would have produced. The response contains no warning and no inconsistency a reader could notice. Unless the answer is checked against a solver that knows which convention was requested, the substitution is invisible.

This paper names that behavior, measures it at scale, identifies three factors that govern it, and tests two mitigations that mostly fail to fix it. Throughout, we use one term for the central phenomenon, defined precisely in Section 2: a model *reverts* on a variable when its answer has the correct magnitude but carries the sign of the training-default convention rather than the instructed one. Reversion is a compliance failure that standard evaluations cannot see, because they either grade magnitudes, accept answers up to sign, or use language-model judges that share the same training priors.

Sign conventions turn out to be an unusually clean instrument for separating *competence* (can the model solve the problem?) from *compliance* (does the instruction win when it conflicts with training?). Three properties make them so. They are *arbitrary*: both choices are physically valid, so the instruction is never a trick. They are *binary*: on any single variable, compliance is a sign. And they are *exactly gradable*: a conventional solver can compute the correct answer under either convention, so no judgment call, rubric, or judge model is involved. Because magnitude and sign are graded separately, a single response yields independent measurements of both axes.

**What we find.** Our results are organized as four claims. (1) Reversion is *silent and directed* (Section 3): sign errors almost never land on arbitrary values; they land on the trained convention’s value, variable by variable, in 99.87% of 58,244 graded cases. (2) Reversion is governed by three measurable factors (Section 4): how the convention’s *reversal* is represented in pedagogy, how hard the problem is, and how capable the model is. (3) Mitigation is conditional at best (Section 5): reasoning modes buy large competence gains and, on most conventions, no compliance; worked in-context examples buy almost nothing anywhere. (4) The framework is falsifiable, and was falsified twice (Section 6): two predictions recorded in the project’s version history before data collection came out opposite to what we predicted, and both refutations are reported and interpreted.

**Why it matters.** A sign is the difference between tension and compression, charging and discharging, stable and unstable. The failure mode we document is worst-case shaped: it is invisible in the answer text, it points consistently toward the training distribution, it grows when problems get harder, and it resists the first mitigation a practitioner would try. Evaluations that do not grade signs against the instructed frame will report such a model as excellent.

## 2 DEFINITIONS AND INSTRUMENT

### 2.1 CONVENTIONS, CONTRACTS, AND CELLS

A *convention* is a sign or orientation choice that is fixed by custom rather than physics. A *convention contract* is the block of an instance’s prompt that states which conventions the answer must use. Every instance in our benchmark states its contract explicitly, including the default one; models are never expected to guess. A *cell* is a version of the same underlying physics problem posed under a particular contract. The *default cell* states the textbook conventions. A *flip cell* changes exactly one convention and leaves everything else identical. Because the physics is shared, a model’s behavior on the default cell and a flip cell of the same problem forms a matched pair, and causal claims about the flipped convention come from within-problem comparisons rather than across-problem averages.

A concrete example makes the design tangible. One circuits instance shows a five-element network and states, among other conventions, “mesh currents ( $i_1, i_2, i_3$ ) are defined CLOCKWISE” (the default cell) or “... defined COUNTER-CLOCKWISE, i.e., the reverse of the usual clockwise choice” (the flip cell). The requested output is a fixed line of `name=value` pairs. Under the flip, the correct values of  $i_1, i_2, i_3$  are the negations of their default-cell values; every other requested quantity is unchanged. A model that answers the flip cell with the default cell’s signs has followed its training rather than its instructions, and the grader can detect this exactly.

### 2.2 REVERSION, PRECISELY

For every instance we compute two reference answers with conventional solvers: the *instructed answer* (correct under the stated contract) and the *default answer* (correct under the textbook contract). A variable is *diagnostic* for a cell when these two answers differ in sign; only diagnostic variables can reveal convention behavior, and all rates in this paper are computed over them. A model’s value for a diagnostic variable is graded into exactly one of four categories: *compliant* (matches the instructed answer within a 5% relative tolerance), *reverted* (magnitude correct but the value matches the *default answer*), *incoherent* (magnitude correct but the sign matches neither reference), or *magnitude-wrong* (a competence error, excluded from compliance rates). The *reversion rate* of a model on a cell is the fraction of its magnitude-correct diagnostic variables that are reverted. Conditioning on magnitude-correctness is what separates the axes: a model can only revert on a variable it was competent enough to compute.

### 2.3 THREE DOMAINS, EIGHT CONVENTIONS

*Circuits*: linear DC networks across seven topology families; requested variables are mesh currents, node voltages, and branch quantities. Its three conventions are mesh direction (default clockwise), the passive sign convention, and the reference node. *Contour*: line integrals of polynomial vector fields around lattice polygons; requested variables are per-edge work integrals, the total circulation, and the flux. Its conventions are traversal orientation (default counter-clockwise), the sign of reported work, and the flux normal. *Vectors*: multi-body torque, angular-momentum, and magnetic-force problems in three dimensions. Its conventions are cross-product handedness (default right-hand rule) and the Newton’s-third-law reporting choice. Difficulty in every domain is a generator setting (element count, polynomial degree, coordinate size), and each domain ships an easy and a hardened tier.

Two design choices matter for later sections. First, the contour cells for clockwise orientation (*cw*) and for reporting work done *against* the field (*wrk*) demand the *byte-identical* output transform: negate every work value. They differ only in which convention is named. Any behavioral difference between them therefore isolates the identity of the convention with the output demand perfectly matched; we use this as a control against “models just get sloppy under load.” Second, each domain also poses every problem under two required solution procedures (mesh vs. nodal analysis; direct integration vs. Green’s theorem; direct summation vs. the transfer theorem), which lets us check that reversion is a property of the *reported frame*, not of any particular solving procedure.

### 2.4 VERIFICATION AND GRADING

Every accepted instance is solved by four independent routes before any model sees it: in circuits, two independently hand-derived formulations, an exact symbolic modified-nodal-analysis solver, and the NGSPICE simulator; in contour, symbolic integration of each parametrized edge, Green’s-theorem double integration over an exact triangulation, an independently implemented rational-arithmetic engine, and numerical quadrature; in vectors, textbook component formulas, Levi-Civita summation, symbolic cross products, and floating-point evaluation, together with exact transfer-theorem identities. Three of the four routes use exact rational arithmetic and must agree to the bit; instances are rejected otherwise, so the released datasets contain zero solver disagreements. The instructed answers for every flip cell are additionally validated by *independent physical realization* rather than by re-applying the same transform: the counter-clockwise circuit cell by re-grounded re-simulation, the left-handed cell by re-solving the entire problem with the Levi-Civita symbol negated, the reaction cell by negating the force sources (125/125 problems pass in each domain). Grading is a deterministic parser over a mandated answer line. No language model participates in generation, verification, grading, or analysis.

### 2.5 MODEL PANEL

We evaluate fifteen configurations: GPT-5 at low reasoning effort, gpt-oss-120b, Gemma-4-31B, DeepSeek-V3, Llama-3.3-70B, Qwen2.5-72B, and the Qwen3 dense family (1.7B, 4B, 8B, 14B, 32B), each Qwen3 size in both its reasoning and non-reasoning mode, which toggles behavior while holding weights fixed. Whole-problem competence spans roughly 790× across the panel. Decoding settings are uniform; serving stack and quantization are pinned and recorded per response. We refer to GPT-5, gpt-oss-120b, and Gemma-4-31B as the *frontier* models: they are the only panel members that solve the majority of problems in every domain, and several claims are therefore stated for them separately.

## 3 REVERSION IS SILENT AND DIRECTED

Our first result is about the *structure* of sign errors, and it holds everywhere we look. Across all domains, tiers, cells, and models, we grade 58,244 magnitude-correct diagnostic variables. If sign errors were noise, errors would scatter between the default frame and incoherent values. They do not: 99.87% of all sign errors equal the trained convention’s value exactly. The 74 incoherent cases (0.15%) all occur in a single place, the circuits reference-node cell, which is the one convention in the study whose transform is affine (subtract a reference potential) rather than a sign flip; a model that

Table 1: Frontier reversion at low difficulty for all eight conventions, by reversal-pedagogy tier (0 = reversal taught as an operation, 1 = taught as an error contrast, 2 = absent from pedagogy). Pooled over GPT-5, gpt-oss-120b, and Gemma-4-31B, with Wilson 95% confidence intervals.

| Convention (domain)       | Tier | Reversion % | 95% CI       | <i>n</i> vars |
|---------------------------|------|-------------|--------------|---------------|
| mesh direction (circuits) | 2    | 76.0        | [73.3, 78.5] | 1,045         |
| passive sign (circuits)   | 1    | 6.1         | [4.6, 8.0]   | 755           |
| reference node (circuits) | 1    | 0.5         | [0.1, 1.7]   | 437           |
| handedness (vectors)      | 1    | 0.2         | [0.1, 0.4]   | 3,698         |
| orientation (contour)     | 0    | 2.3         | [1.9, 2.8]   | 3,902         |
| reaction pair (vectors)   | 0    | 1.4         | [1.1, 1.9]   | 3,661         |
| flux normal (contour)     | 0    | 0.1         | [0.0, 0.8]   | 736           |
| work sign (contour)       | 0    | 0.0         | [0.0, 0.1]   | 4,023         |

applies the shift partially lands between frames. On all seven pure sign-flip conventions, spanning tens of thousands of variables, the incoherence count is exactly zero. When these models get a sign wrong, they are not guessing. They are reproducing their training distribution, one variable at a time.

The matched-pair design makes the effect causal rather than correlational. In circuits, posing the same problem under the counter-clockwise contract instead of the default breaks 401 previously-solved problems across the panel while fixing 6; exact McNemar tests on these within-problem pairs are significant at  $q_{BH} \leq 0.006$  in every competent family. Competence does not protect a model: Gemma-4-31B solves 85% of circuit magnitudes and still reverts on 77.9% of its diagnostic signs. Nor is this a general inability to follow unusual instructions: the same models comply at 85–97% with matched instructions that are equally unusual but touch no convention (e.g., reporting variables in reverse order), and re-sampling the entire measurement five times moves the headline rates by only a few points.

## 4 WHAT GOVERNS REVERSION

### 4.1 HOW THE REVERSAL IS TAUGHT

The eight conventions we instruct differ sharply in how their *reversal* is represented in the material models learn from, and this, more than anything else, predicts what models do. We code each convention on a three-level ordinal scale by auditing standard textbooks (the audited list and coding criteria are in Appendix B). *Tier 0*: the reversal is taught as a routine operation. Reversing a contour’s orientation negates the integral; this is a theorem statement in every calculus text, with exercises. Reaction forces and inward normals are likewise curriculum material. *Tier 1*: the reversal appears only as an error contrast or a partial treatment; every introductory physics text warns that using the left hand gives the opposite direction, which teaches the flip while forbidding it. *Tier 2*: the reversal is essentially absent. We could not find a single worked counter-clockwise mesh example in any circuits text we audited, and our own search for a second tier-2 convention failed (Section 6).

Table 1 and Figure 1 show frontier reversion at low difficulty against this scale. The separation is stark. Tier-0 conventions are obeyed at 2.3–1.4%. The single tier-2 convention is disobeyed at 76.0%. Tier-1 conventions fall between or split by model. A weighted logistic regression of reversion on tier across all eight conventions yields  $z = 45.4$ . We emphasize what this gradient means: the models are not incapable of sign-flipped reporting, since they perform the identical transform when the convention naming it is one whose reversal they have seen taught. What they lack is any trained representation of *this* reversal, and no instruction we found supplies it at inference time (Section 5).

### 4.2 HOW HARD THE PROBLEM IS

Obedience on taught-reversal conventions is not unconditional; it has a load limit. Within the contour domain, we harden the problems (higher polynomial degree, more polygon edges) and measure reversion as each model’s competence strain rises, where strain is simply the fraction of default-cell problems whose magnitudes the model now gets wrong. Reversion on the orientation convention

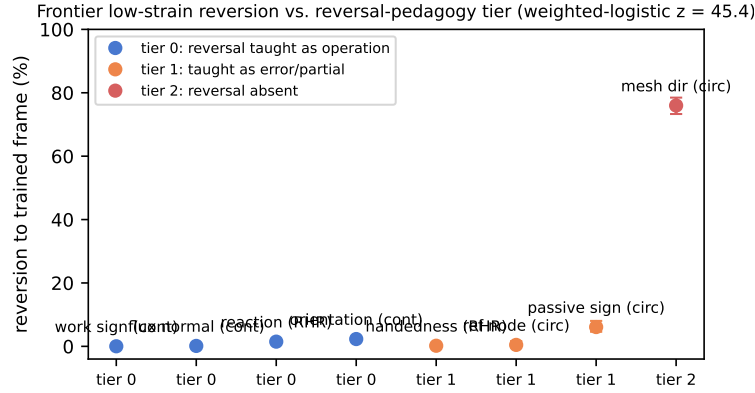
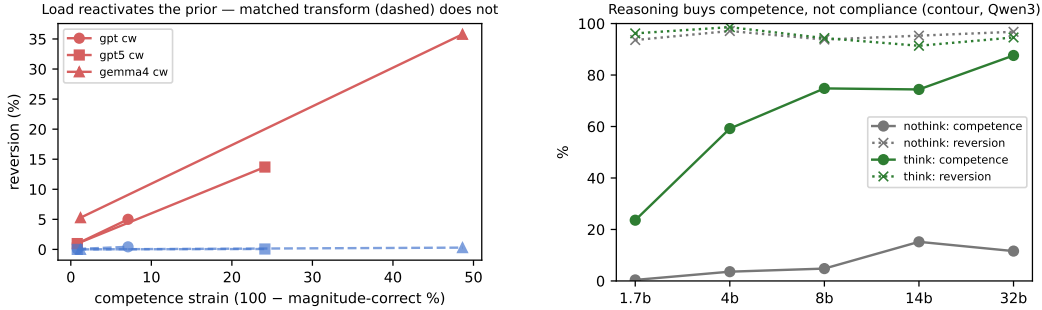


Figure 1: Reversion against reversal-pedagogy tier across all eight conventions.

Figure 2: Left: reversion against competence strain in the contour domain; dashed lines are the transform-matched `wrk` control. Right: the Qwen3 reasoning toggle raises competence at every scale while orientation reversion stays above 90%.

climbs monotonically: gpt-oss from 1.0% to 5.0% at 7% strain, GPT-5 from 1.0% to 13.7% at 24%, and Gemma from 5.3% to 35.8% at 49% (a five-fold resampling of this last, largest figure gives mean 34.4%, SD 6.4 points). The crucial comparison is the `wrk` control cell, which demands the identical negation of the same variables: it stays below 1% for every frontier model at every difficulty (Figure 2; weighted logistic  $z = 18.1$ ). Difficulty does not cause generic instruction-forgetting. It selectively reactivates the trained default of the specific convention under pressure. One consequence is worth stating plainly: the failure is worst exactly where users need the model most, on hard problems. (Rows lost to truncation in the hardened tier are the most-strained instances, so observed rates are, if anything, underestimates.)

#### 4.3 HOW CAPABLE THE MODEL IS

The pedagogy gradient of Section 4.1 describes frontier models. Below the frontier, compliance degrades even on tier-0 conventions: the Qwen3 family reverts on contour orientation at 91–99% at every scale and in both reasoning modes (Table 2), and DeepSeek-V3, Llama-3.3-70B, and Qwen2.5-72B revert at 92–94%. Even at this floor, convention identity still matters: the same weak models revert far less on the transform-matched `wrk` cell (for example, 92% versus 35% for DeepSeek-V3), so the gradient’s fingerprint is visible below the capability threshold even though absolute compliance is not achievable there. Following a taught reversal under load is, itself, an emergent capability.

Table 2: The Qwen3 reasoning toggle on contour: competence rises at every scale; orientation reversion does not move.

| Scale | no-think |        | think   |        |
|-------|----------|--------|---------|--------|
|       | Comp. %  | Rev. % | Comp. % | Rev. % |
| 1.7b  | 0        | 93.6   | 24      | 96.2   |
| 4b    | 4        | 97.1   | 59      | 98.6   |
| 8b    | 5        | 93.8   | 75      | 94.3   |
| 14b   | 15       | 95.3   | 74      | 91.4   |
| 32b   | 12       | 96.8   | 88      | 94.6   |

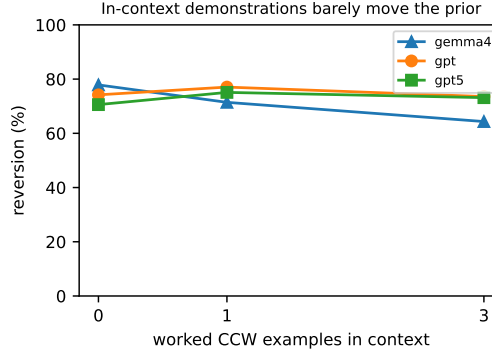


Figure 3: Circuits reversion against the number of in-context worked counter-clockwise examples.

## 5 WHAT MITIGATION DOES AND DOES NOT DO

### 5.1 REASONING MODES: LARGE COMPETENCE GAINS, CONDITIONAL COMPLIANCE

The Qwen3 family can run with its reasoning mode on or off at identical weights, which isolates the effect of test-time deliberation. On contour, the toggle raises magnitude-competence from 0–15% to 24–88% across five scales while moving orientation reversion by two to four points (Table 2); the same dissociation holds in circuits. Deliberation, by itself, does not make the instruction win.

The vectors domain, however, reveals the boundary of this negative result, and we report it because it sharpens rather than weakens the claim. On the handedness convention, reasoning *does* restore compliance: reversion falls from 96.0% to 2.9% at 4B and to 0.0% at 8B (Table 3, Appendix A). In the same responses, the reaction-reporting convention keeps slipping at 32.1–35.9%. The pattern across all three domains is consistent: test-time deliberation can *retrieve* a reversal that is taught and procedurally shallow, a per-operation sign rule the chain of thought can state and apply each time it is needed. It does not conjure a reversal that was never taught (circuits), it does not sustain one that must be maintained across every step of a long derivation (contour orientation under load), and it does not reliably hold a framing convention that applies once at reporting time (reaction). What compute buys is retrieval, not override.

### 5.2 WORKED DEMONSTRATIONS: ALMOST NOTHING

If reversion reflects a missing training representation, perhaps supplying that representation in-context cures it. We prepend one or three fully worked, solver-verified examples of counter-clockwise mesh analysis (held-out problems, different topologies) to the same circuit instances. The result is a nearly flat dose-response (Figure 3): Gemma declines modestly (77.9%→71.4%→64.3%, about 4.5 points per example), while gpt-oss (74.2%→73.6%) and GPT-5 (70.6%→75.1%→73.2%) do not move at all. Reading a correct demonstration of exactly the required behavior, immediately before the problem, does not override the prior. The mitigation practitioners would reach for first is the one we can most confidently say does not work.

## 6 TWO PRE-REGISTERED PREDICTIONS, BOTH REFUTED

Before collecting the corresponding data, we committed two predictions to the project’s version history. **Prediction 1:** left-handed cross products would behave like counter-clockwise mesh currents, an untrained reversal, and frontier models would revert heavily. *Refuted:* across three difficulty tiers and roughly 3,700 diagnostic variables, frontier reversion on the left-hand rule is 0.0–0.6% (0/1231 for Gemma at 95% competence). In hindsight the pedagogy explains the miss: every textbook teaches the left-hand outcome as an explicit warning, which is training signal for the flip. This refutation is what forced the ordinal coding of Section 4.1, and our subsequent failure to find any second tier-2 convention is itself evidence for how unusual the mesh convention’s pedagogical silence is. **Prediction 2:** in-context demonstrations would collapse reversion if it stems from a missing representation. *Refuted,* as Section 5 shows. We report both because a measurement framework that only ever confirms its authors’ expectations is not measuring anything; the claims that survive are the ones these refutations disciplined.

## 7 RELATED WORK

Instruction-following benchmarks grade compliance with formatting and content constraints (Zhou et al., 2023); counter-default instructions have been studied as inverse instructions (Stephan et al., 2025) and counterfactual task variants (Wu et al., 2024). Relative to these, our contribution is machinery that separates compliance from competence *within* a single response using sign-exact oracles, plus controls (transform-matched cells, procedure contracts) that localize the failure to convention identity. Sycophancy work documents deference to user framing (Sharma et al., 2024); reversion persists with no user stance to defer to. Test-time-scaling studies measure capability gains from deliberation (Muennighoff et al., 2025; DeepSeek-AI, 2025); we show those gains transfer to compliance only under narrow conditions. The benchmark extends a published single-domain probe (Ravishankara, 2026a) and follows deterministic evaluation practice from engineering-plot benchmarks (Ravishankara, 2026b); canary-based contamination protection follows Srivastava et al. (2022); statistics use exact McNemar tests (McNemar, 1947) and Wilson intervals (Wilson, 1927).

## 8 DISCUSSION

**Implications for evaluation.** Any benchmark that grades magnitudes, accepts answers modulo sign, or scores with a language-model judge will award full credit to every reverted answer in this study. Sign-exact grading against the instructed frame is cheap when the domain permits it, and our results suggest it measures a dimension of controllability that accuracy does not.

**Implications for deployment.** The failure mode is silent, directed toward the training distribution, load-sensitive, and robust to prompting. In engineering use, a deterministic frame-check layer (recompute the requested convention’s signs from the model’s own magnitudes) costs almost nothing and, on this evidence, is worth more than any reasoning budget.

**Limitations.** The tier-2 pole of the pedagogy scale is occupied by a single convention; we searched for a second and found only a tier-1 case, so we present a measured gradient rather than a two-point law. Difficulty could not be raised enough in the vectors domain to strain frontier models, so the taught-reversal  $\times$  high-load cell is measured only in contour. Pedagogy tiers are coded from a finite audit of standard textbooks (Appendix B). Our analysis is behavioral; we do not localize the mechanism inside the network. Serving stacks are mixed across the panel, though pinned and recorded, and hardened-tier truncation censors the most-strained rows in the conservative direction.

**Release.** The generators, oracles, validators, grading, and analysis code are released publicly; every dataset regenerates byte-identically from a seed, so the public artifact is a one-command generator rather than a crawlable answer key. Frozen instances and model outputs are available under gated access with embedded canaries. We take this structure seriously because this benchmark measures a training-frequency effect: if its instances and instructed answers entered a future training corpus, they would not merely inflate scores, they would administer the treatment and invert the measurement.

## REFERENCES

- DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv:2501.12948*, 2025.
- Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- Niklas Muennighoff et al. s1: Simple test-time scaling. In *arXiv:2501.19393*, 2025.
- M. Ravishankara. Circuchain: Disentangling competence and compliance in LLM circuit analysis. In *Proc. IEEE SoutheastCon*, 2026a. *arXiv:2602.15037*.
- M. Ravishankara. Plotchain: Deterministic checkpointed evaluation of multimodal LLMs on engineering plot reading. In *Proc. IEEE SoutheastCon*, 2026b. *arXiv:2602.13232*.
- Mrinank Sharma et al. Towards understanding sycophancy in language models. In *Proc. ICLR*, 2024.
- Aarohi Srivastava et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *arXiv:2206.04615*, 2022.
- M. Stephan et al. Inverse IFEval: Can LLMs unlearn stubborn training conventions to follow real instructions? In *arXiv:2509.04292*, 2025.
- Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *JASA*, 22: 209–212, 1927.
- Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, and Yoon Kim. Reasoning or reciting? exploring the capabilities and limitations of language models through counterfactual tasks. In *Proc. NAACL*, 2024.
- Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models. In *arXiv:2311.07911*, 2023.

## A REASONING TOGGLE ON THE VECTORS DOMAIN

Table 3: Qwen3 reasoning toggle on the vectors domain (hardened tier): competence % and reversion % per convention. Reasoning collapses the taught, shallow handedness reversal but not the reaction-reporting convention.

| Scale | Comp. | no-think |          | Comp. | think   |          |
|-------|-------|----------|----------|-------|---------|----------|
|       |       | lh rev.  | rxn rev. |       | lh rev. | rxn rev. |
| 1.7b  | 16    | 89.2     | 93.0     | 37    | 35.9    | 83.4     |
| 4b    | 24    | 96.0     | 92.5     | 72    | 2.9     | 42.1     |
| 8b    | 23    | 29.7     | 22.1     | 79    | 0.0     | 32.1     |

## B TEXTBOOK AUDIT BEHIND THE PEDAGOGY TIERS

For each convention we examined the treatment of its *reversal* in standard textbooks: for circuits, Nilsson & Riedel; Hayt, Kemmerly & Durbin; Irwin & Nelms; Sadiku; and Kuphaldt’s open text; for multivariable calculus, Stewart; Thomas; Marsden & Tromba; and the OpenStax Calculus series; for mechanics and electromagnetism, Halliday, Resnick & Walker; Serway & Jewett; Young & Freedman; Griffiths; and OpenStax University Physics. A convention’s reversal is coded *tier 0* if at least one audited text states it as a rule or theorem and at least one provides exercises using it (orientation reversal, reaction pairs, inward normals, work against a field); *tier 1* if audited texts mention the reversal only as an error contrast or partial treatment without worked examples (left-hand rule warnings, active sign convention, non-ground references); *tier 2* if no audited text provides

any worked treatment of the reversal (counter-clockwise mesh currents; we found none). The coding predates the vectors-domain results (Section 6 describes how a mis-anticipation of the left-hand rule’s tier motivated this audit).

## C REPRODUCIBILITY

All datasets are deterministic functions of a configuration file and a seed; regeneration commands, manifests with canary identifiers, transform-validation reports (125/125 per domain), the complete statistical freeze backing every number in this paper, and per-response provenance (model identifier, serving stack, quantization, sampling parameters) accompany the code. No language model is involved at any point in generation, verification, grading, or analysis.